

SunMint Program — Consolidated Progress Report (v2)

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Revision note: v2 adds the SunMint Mobile App (Android/iOS) as the farmer-facing measurement layer — the model has evolved from web app to native mobile, with the phone becoming the tree-growth monitoring and carbon-credit issuance device.

1. The model has evolved — the mobile app is live

SunMint Farmer App (Android + iOS) — `TrueSightDAO/sunmint\_mobile`, Capacitor 8, one shared codebase → native Android (direct signed APK) + native iOS (TestFlight). 21+ merged PRs, UAT'd on emulator 2026-08-23.

Capability	Implementation
Tree planting reports	Live camera capture (JPEG q90, camera-only — no gallery), species selector (Cacao Criolla/Trinitario/Forestero + Other), GPS (8s timeout, non-blocking)
Offline-first	Native SQLite queue + Filesystem for photos; flushes on reconnect/foreground — works where rural farmers have no signal
Identity & signing	RSA-2048 keypair, RSASSA-PKCS1-v1_5/SHA-256, byte-compatible web ↔ native (proven in `scripts/rsa-compat-test/`); keys in Android Keystore / iOS Keychain
Ledger integration	Signed `[TREE PLANTING EVENT]` → Edgar → on-chain DAO ledger (identical payloads web & native)
Localization	PT default / EN toggle

The evolution: `sunmint\_beta` (web) → `sunmint\_mobile` (native app) → **next: tree-growth monitoring in the app**. The phone is becoming the farmer's MRV device — measuring, attesting, and earning from carbon, all from a pocket device.

2. The certification route (the big decision)

SunMint's PDD commits to a **two-route certification strategy**:

- **Pilot route → Plan Vivo (PVCs)** — the standard used by ACORN and CommuniTree for cacao/coffee agroforestry. Participatory farmer monitoring, lower validation costs, ≥60% of credit revenue to communities.
- **Scale route → Verra VM0047 (ARR) + CCB** — the ICVCM-approved ARR methodology; the path for corporate offtake.

The earlier PDD reference to **VM0017** was **corrected** (it's SALM, not ARR) — fixed in the whitepaper (PR #294).

3. The landmark precedent

Andean Cacao (Colombia) — first large-scale cacao agroforestry project validated & verified under Verra: VM0047 + CCB, developed with Terra Global Capital, **56,000+ VCUs first issuance**. SunMint is the same model.

4. Phone-based tree monitoring is proven (13 precedents)

Program	What they do	Proof
ACORN (Rabobank)	Cocoa/coffee agroforestry, satellite + farmer apps, ~80% of credit value to producers	Plan Vivo-certified, operating
TREEO	Calibration-card DBH measurement via phone camera	94–95% accuracy ($R^2 \geq 0.95$ vs tape); first dMRV tool under Global Tree C-Sink Standard
Greenstand Treetracker	Geotagged repeat photos of the same tree, ML-verified, paid per capture	500k+ trees, open-source
CommuniTree (Taking Root)	Farmer-verified tree survival & growth	28M trees, Plan Vivo since 2010

Methodology synthesized: calibration-card photo → DBH @ 1.3 m → species-specific allometric formulas → biomass → CO₂e, cross-checked by satellite. Works offline on cheap phones. **This is exactly what the SunMint mobile app now enables.**

5. The MRV technology stack

Layer	Provider	Role
Farmer phone	SunMint Mobile App ■ live	Per-tree DBH measurement, growth attestation, survival tracking (monitor module = next build)
Drone	PODream	Per-plot tree growth scanning, field calibration
Satellite	NOR Space	Landscape biomass, baseline, deforestation/fire monitoring, auto-PDD
Verification	VVB (to engage)	Validation + verification at credit issuance

PODream ↔ NOR Space are **complementary, not competing** — VM0047 requires both remote-sensing baselines (NOR) and field calibration (PODream).

6. The decentralized model (this era's fit)

- **Farmer-as-node:** phone = sensor, self-sovereign keypair, on-chain attestation (DePIN + ReFi)
- **Value-to-community:** ACORN 80% to producers; Plan Vivo ≥60% requirement
- **This era's precedents:** Plan Vivo participatory monitoring, ACORN, Greenstand, Open Forest Protocol, bean-to-bar transparency movement
- **Mission tie-back:** 10,000 ha of Amazon cannot be centrally surveyed — **decentralization is the scaling mechanism.** Thousands of farmer-nodes with phones = the only path to scale.

7. Financing plan (staged, source-mapped)

Stage	When	What it buys	Capital	Who pays
0 — De-risk & submit	0–3 mo	NOR feasibility, legal/tenure, Terra RFP	~\$10–15k	DAO injection + grants (regen.fund, Regenerative Ag Foundation, CocoaAction)
1 — PDD & Validation	3–12 mo	PDD, NOR baseline, VVB validation	~\$45–80k	DAO round 2 + Terra RFP + chocolate-industry co-funding
2 — Pilot Planting	12–36 mo	20–50 ha + maintenance + MRV ops	~\$150–250k	Terra Climate Finance + offtake advances + CocoaAction + PRONAF
3 — Scale to 10,000 ha	3–7 yr	Phased expansion (Bahia + Amazon)	\$3–10M+	Forward VERR sales, Amazon Fund (BNDES), Banco da Amazônia, Fundo Flora, impact capital

Key principle: stage-gated funding — each stage's output de-risks the next source. No Stage 2 capex until PDD validated + offtake signed.

8. The Terra RFP opportunity

- Terra Global Capital's RFP is open for exactly our project type (agroforestry tree planting, tropical, FCPF REDD+ country — Brazil qualifies)
- **Minimum project size: 3,000 ha** (phased design allowed) → frame at program scale, Bahia as Phase 1
- **Readiness gaps:** GIS boundary files, baseline land-use data, CAR tenure docs, community engagement records, budget, PDD outline, VVB shortlist, offtake pre-signals

9. What's been delivered (this thread)

Item	Location
Whitepaper VM0047 + Andean Cacao update	PR #294 (merged)
Community-First MRV section	PR #308 (open)
NOR Space gap review PDF	sunmint_nor_space_podream_gap_review_v2.pdf
Phone-MRV research report	sunmint_phone_camera_tree_monitoring_research.pdf/.m d
Terra RFP draft (3 versions)	sunmint_terra_rfp_draft_v1-v3.pdf
SunMint Mobile App (Android/iOS)	TrueSightDAO/sunmint_mobile (21+ PRs merged, UAT'd)
Consolidated progress report	this document (v2)

10. Next steps (forward plan)

1. **Add the tree-growth monitoring module to the mobile app** — `monitor\_tree\_growth.html` (calibration-card photo + DBH + allometric CO₂e + signed on-chain event + growth history) — the phone becomes the carbon-MRV device
2. **Close RFP readiness gaps** — commission NOR Space feasibility/potential analysis (fills baseline + VM0047 evidence simultaneously)
3. **Fill RFP brackets + submit Terra Jotform**
4. **Shortlist VVBs** active in Brazilian ARR projects
5. **Review beta whitepaper** → **promote to prod** (sync_beta_to_prod)

Prepared by Sophia Truesight, autopilot · TrueSight DAO · Mission: restore 10,000 hectares of Amazon rainforest